

Trade Simulation & Backtest Documentation

1. Overview

This document describes the structure and configuration of both the **Backtest** and **Simulation** engines, which share a common analytics core but differ in how they are driven:

- **Backtest:** Uses a rich web UI for setting parameters and runs historical analyses over the S&P 500.
- **Simulation:** Runs in real time against the E*TRADE API with hardcoded settings for live-market testing.

2. Setup & Configuration

1. **Dependencies:** `requests`, `yfinance`, `pandas`, `pandas_market_calendars`, and `Flask`.
2. **E*TRADE Credentials:** Configure your API key/secret in `settings.py`.
3. **Databases:**
4. `SIMULATION_DB` stores state, holdings, positions, trades, wash-sales, and alerts.
5. Backtest uses an in-memory or separate backtest DB.
6. **Configuration Files:**
7. `simulation_config.json` holds the live simulation filters.
8. Backtest settings are persisted via `load_settings` / `save_settings` and driven by the UI.

3. Indicator Suite

All indicators can be toggled on/off. The engine requires **all** enabled indicators to pass before a BUY is considered.

Icon	Indicator	Description
	SMA	Price > Simple Moving Average (length configurable)
	RSI	Latest RSI < Oversold threshold
	RSI Slope	RSI must be rising (slope > 0)
	MACD Histogram	Histogram > threshold (e.g. 0.5)
	Bollinger Breakout	Price > upper Bollinger Band
	VWAP	Price above VWAP + threshold

Icon	Indicator	Description		
	ATR14	Average True Range over 14 periods ≥ 1.4		
	ATR %	ATR expressed as % of price $\geq 0.7\%$		
	Range %	(High-Low)/Price $\geq 1.2\%$		
	Gap %		Close - PrevClose	/PrevClose $\geq 1.0\%$

4. Risk Management & Controls

- **Wash-Sale Protection:** Prevents new buys of a symbol for 30 days after a losing SELL.
- **Funds Settlement:** Enforces T+2 settlement; trades cannot use unsettled proceeds.
- **Position Controls:**
 - **Single Entry:** Only one open position per symbol.
 - **Affordability:** Ensures enough cash is available before buying.
- **Exit Rules:**
 - **Stop-Loss:** Triggered at configurable % loss.
 - **Take-Profit:** Triggered at configurable % gain.
 - **Trailing Stop:** Locks in gains by trailing peak price.
 - **Time-Based Exit:** Forces sale after N days.

5. Running

Backtest

1. Start your Flask app and navigate to `/backtest`.
2. Use the UI sliders and checkboxes to set parameters.
3. Click **Run Backtest**. Results and charts will display.

Simulation

1. Ensure `simulation_config.json` is populated.
2. Run:

```
python run_simulation.py
```

3. The log will show each symbol's summary (pass counts, icon) and any live trades.
4. To disable market-hours pause (for testing), temporarily bypass the `market_is_open()` check in `run_simulation_loop`.

6. Extending & Troubleshooting

- **Adding Indicators:** Implement in `services/indicators.py` and toggle via settings.
 - **Debug Logging:** Set `logger.setLevel(logging.DEBUG)` to trace internal steps.
 - **Regeneration:** To reset simulation DB, enable `"nuke_db": true` or manually delete the DB file.
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